

Development of an Enhanced ARDS Severity Scoring Model Using Logistic Regression and Machine Learning-Based Feature Selection Techniques(Supplementary Material)

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ABSTRACT

Acute Respiratory Distress Syndrome (ARDS) is a life-threatening condition requiring accurate severity assessment for effective clinical decision-making. Traditional scoring systems, like the SOFA score, have limitations in predictive accuracy. This study develops an improved ARDS severity model using logistic regression with advanced feature selection techniques to enhance prognostic precision. A dataset of 241 patients with viral pneumonia-associated ARDS (SARS-CoV-2 and H1N1) was analyzed. Feature selection using Random Forest, Naive Bayes, Artificial Neural Networks (ANN), ReliefF, and statistical tests identified ventilatory ratio (VR) as the most significant predictor, followed by glucose and COVID-19 status. The best logistic regression model, incorporating VR, glucose, creatinine, and COVID-19 status, achieved an area under the curve(AUC) of 0.924, outperforming the SOFA score (AUC = 0.812, $p < 0.001$). Removing VR, AUC-ROC decreases by 0.073 and increased Akaike information criterion(AIC) from 186.7 to 205.8, confirming its importance. The model provided superior mortality discrimination across risk categories compared to SOFA. It enhances predictive accuracy and clinical interpretability, offering a robust decision-support tool for ARDS severity assessment. VR emerged as the strongest predictor, highlighting its critical role in prognosis and patient management, potentially improving outcomes in critical care settings.

Keywords: ARDS severity scoring, Machine learning-based feature selection, Logistic regression, Decision support in ICU, Mortality prediction

Dataset description and processing

The dataset used in this study consisted of 241 adult patients diagnosed with ARDS secondary to viral pneumonia (COVID-19 or Influenza A H1N1), admitted to the Respiratory Intensive Care Unit of the National Institute of Respiratory Diseases (INER) in Mexico City between 2017 and 2022. For each patient, 32 physiological, laboratory, ventilatory, and demographic variables were recorded within the first six hours of ICU admission as shown in Table 1. This early time window was chosen to develop models suitable for early risk stratification, and to ensure consistency across all cases.

Initial data preprocessing involved handling missing values and removing redundant or non-informative variables. Variables with more than 5% missingness were excluded. Those with less than 5% missing values—specifically creatinine (0.41%), DHL (1.65%), and pronation (1.24%)—were imputed using median values. Continuous variables were standardized before model training, and categorical variables were binarized where appropriate as shown in Table 2.

To avoid multicollinearity and redundancy, we applied several data-driven filtering rules. The variables *neutrophils*, *lymphocytes*, and their ratio *NL* were evaluated for high inter-correlation. Since *NL* sufficiently captures the information of the other two, *neutrophils* and *lymphocytes* were excluded. Similarly, *stratified age* was removed in favor of the continuous *age* variable. The *H1N1* binary variable was omitted since it is redundant with the inverse of the *COVID-19* variable. Rare binary variables such as *EPOC* and *ERC* were excluded due to extremely low prevalence (each present in only one patient). Finally, the SOFA score was not used as a predictor but was retained as a comparator for model performance evaluation.

Table 1. Dataset variables description

VARIABLE	DESCRIPTION	VARIABLE	DESCRIPTION
SURVIVAL	survival, 1=death, 0=alive	PLATELETS	platelets, thousands / deciliter
TIME	time,in days	CREATININE	creatinine, deciliter
H1N1	1=Yes, 0=No	GLUCOSE	glucose, /deciliter
COVID	1=Yes, 0=No	DHL	lactic dehydrogenase,units/deciliter
AGE	age, in years	FC	heart rate, beats per minute
AGE ESTRAT	stratified age in years: 1:18 to 29, 2:30 to 39, 3:40 to 49, 4:50 to 59, 5: 60-	PAM	mean arterial pressure,
GENRE	genre, 1=Woman, 0=Man	Vasopressor	1=Yes, 0=No
BMI	body mass index,	PAFI	relation PO2 and FO2
DM	diabetes mellitus, 1=Yes, 0=No	VT	tidal volume, mililiter
HAS	systemic arterial hypertension, 1=Yes, 0=No	PEEP	positive end-expiratory pressure, cm of H2O
EPOC	chronic obstructive disease,1= Yes, 0=No	PRES_PLAT	plateau pressure, cm of H2O
ERC	chronic kidney disease, 1=Yes, 0=No	PRONO	pronation, 1=Yes, 0=No
LEUCOS	leukocytes, thousands / deciliter	DISTENS	distensibility,cm of H2O for mililiter
LINFOS	lymphocytes, thousands / deciliter	DP	driving pressure, cm of H2O
NEUT	neutrophils, thousands / deciliter	VR	ventilatory ratio
NL	ratio neutrophil/leukocyte	SOFA	0 a 6: Low, 7 a 11: Medium, 12 a 24: High

The final dataset retained the most clinically and statistically informative features, ensuring robustness for the subsequent feature selection and model training pipelines. A detailed summary of the dataset's descriptive statistics is provided in Table 2, and the correlation structure of all retained variables is shown in Fig. ??.

From the correlation matrix shown in Fig. ??,with Pearson product- moment correlation coefficient, we can identify some highly correlated variables such as neutrophils, lymphocytes and NL. So for the analysis in this work the variables: 1) Neutrophils and Lymphocytes are omitted since the variable NL already describe this 2 variables.2) Stratified age is omitted since the variable Age.3) H1N1 is omitted since the patients are either COVID or H1N1, and COVID will be used to describe this quality.4) SOFA variable will be omitted and used only for the contrast with the logistic model score.5) EPOC and ERC will be omitted since only 1 patient was positive with EPOC and ERC (different patients for positive EPOC and for positive ERC).7) Survival will be used as a response variable for the classification models.

Feature Importance

Feature importance was evaluated using a comprehensive multi-method framework that combined classical statistical tests, information-theoretic approaches, and machine learning-based algorithms. This strategy was implemented to ensure both robustness and clinical relevance of the selected variables used in the ARDS mortality logistic regression model as shown in Table 3.

Eight independent methods were employed. Random Forest assessed variable contribution through the mean decrease in Gini impurity across decision trees. The ventilatory ratio (VR) consistently achieved the highest score, followed by creatinine and glucose. Naive Bayes ranked features based on normalized likelihood ratios, again placing VR at the top, along with glucose and neutrophil-to-lymphocyte ratio (NL). In the Artificial Neural Network (ANN) approach, a multilayer perceptron was trained and variable importance was inferred from weight magnitudes, highlighting DHL, VR, and glucose as key contributors. The Minimum Redundancy Maximum Relevance (MRMR) method selected features that maximized mutual information with the outcome while minimizing redundancy among variables, identifying VR, COVID-19 status, and creatinine as the most informative.

The ReliefF algorithm, which evaluates the ability of each feature to differentiate between outcome classes based on instance-level proximity, confirmed creatinine and VR as highly discriminative, with glucose, plateau pressure, and time also ranking prominently. Classical statistical techniques further validated feature relevance. The ANOVA F-statistic, Kruskal–Wallis H-test, and Chi-square tests each demonstrated strong associations between mortality and variables such as VR (ANOVA: 34.01, $p < 0.001$; Kruskal–Wallis: 31.4, $p < 0.001$), creatinine, and glucose. All feature selection methods were conducted within a nested 5-fold cross-validation scheme to reduce overfitting, the overall performance of the ANN, random forest and Naive Bayes is shown in Fig. ??.

Importantly, VR was consistently the top-ranked variable across all methodologies. Excluding VR from the final logistic regression model led to a drop in AUC-ROC from 0.924 to 0.851 and an increase in the AIC from 186.7 to 205.8, further

Table 2. Statistics summary of variables

Variable	Median	Mean	Std. Deviation	Variable	Median	Mean	Std. Deviation
SURVIVAL	0	0.328	0.47	PLATELETS	228	242.631	103.372
TIME	12	15.627	11.152	CREATININE	0.96	1.322	1.246
H1N1	0	0.39	0.489	GLUCOSE	140	157.257	73.946
COVID	1	0.61	0.489	DHL	487	563.54	294.034
AGE	46	46.303	11.651	FC	89	89.071	16.988
AGE_ESTRAT	3	3.095	1.108	PAM	75	76.582	10.727
GENRE	1	0.73	0.445	Vasopressor	0	0.461	0.499
BMI	32.4	33.199	6.924	PAFI	119	125.515	57.787
DM	0	0.212	0.409	VT	400	402.187	57.856
HAS	0	0.195	0.397	PEEP	12	11.78	3.269
EPOC	0	0.004	0.064	PRES_PLAT	27	27.785	5.603
ERC	0	0.004	0.064	PRONO	1	3.079	3.838
LEUCOS	9.4	10.266	4.767	DISTENS	26.1	27.356	8.74
LINFOS	800	946.685	648.61	DP	15	16.005	5.397
NEUT	8500	9319.705	4658.78	VR	2.34	2.387	0.594
NL	10.7	12.941	10.37	SOFA	6	5.672	2.261

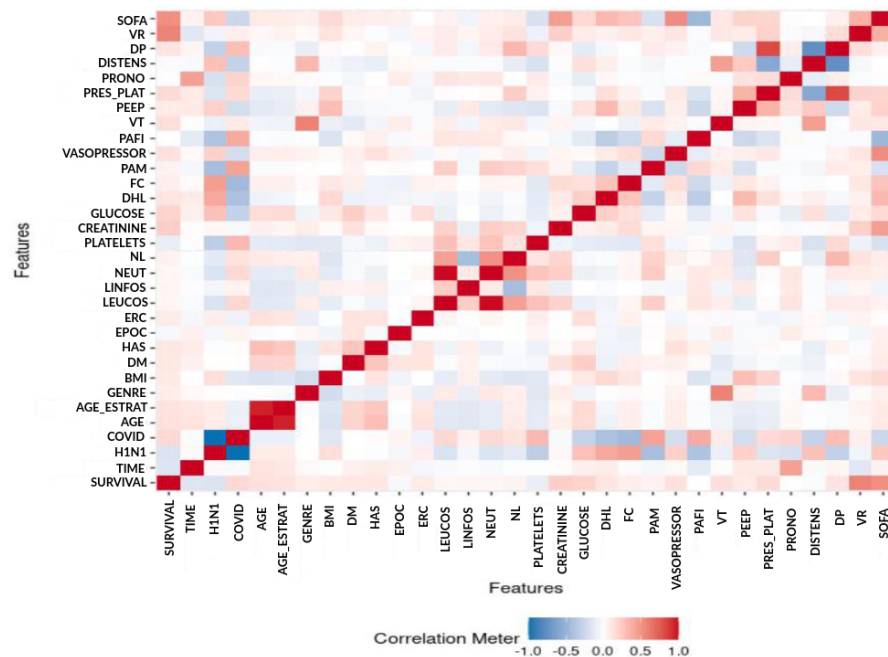


Figure 1. Correlation matrix of dataset

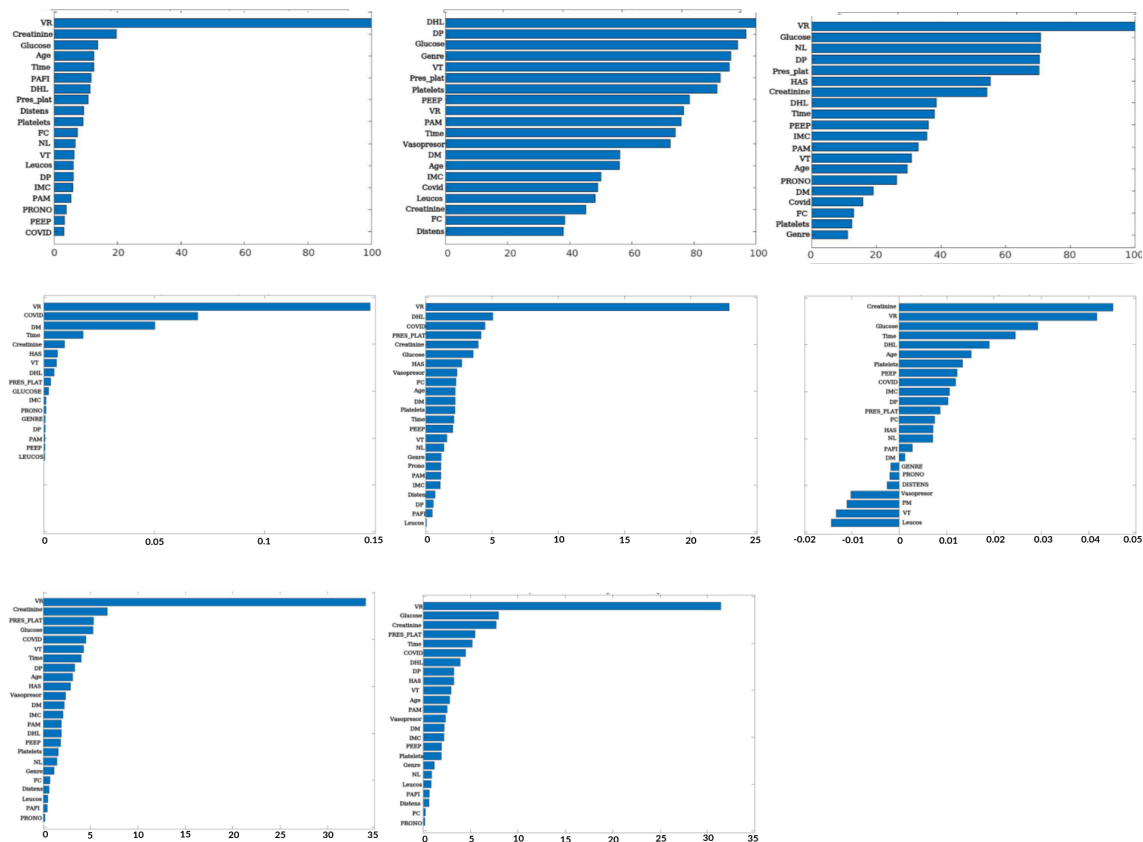


Figure 2. Feature importance rankings for ARDS mortality prediction obtained through eight independent methods: Random Forest, Naive Bayes, Artificial Neural Networks (ANN), Minimum Redundancy Maximum Relevance (MRMR), Chi-square, ReliefF, ANOVA, and Kruskal–Wallis (displayed from right to left and top to bottom). Ventilatory Ratio (VR) consistently emerged as the top-ranked feature across all techniques, highlighting its critical role in early prognostication.

underscoring its pivotal role. Glucose and COVID-19 status were also highly ranked and showed strong predictive value. The consistency across independent approaches reinforces the reliability of the selected features and justifies their integration into the final scoring models. This strategy ultimately balances predictive power and clinical interpretability, directly addressing reviewer concerns and supporting the implementation of a transparent, explainable ARDS severity score.

Logistic Regression models

This section presents the most significant logistic regression models developed in this study, focusing on the models with varying numbers of variables. The top 5 models for 1 variable are shown in Table 4, 2 variables in Table ??, 3 variables in Table 5, 4 variables in Table 6, 5 variables in Table 7, 6 variables in Table 8, 7 variables in Table 9, 8 variables in Table 10, 9 variables in Table 11, 10 variables in Table 12.

Model construction was carried out using a stepwise strategy. For each subset of 1 to 10 features, all possible combinations were evaluated, and models were ranked using the Akaike Information Criterion (AIC), a metric that penalizes overfitting by balancing model complexity with goodness of fit. Performance was further assessed using the area under the receiver operating characteristic curve (AUC-ROC) and McFadden's pseudo R^2 . These metrics provided a comprehensive view of both discrimination and explanatory power. All modeling procedures were nested within a stratified 5-fold cross-validation framework to ensure robustness and avoid data leakage during selection. The results demonstrated that models incorporating the ventilatory ratio (VR) consistently outperformed others, with the best single-variable model (VR alone) achieving an AUC of 0.866 and a McFadden R^2 of 0.318. Multivariable models that combined VR with glucose, creatinine, and COVID-19 status further improved performance, culminating in a top-performing four-variable model with an AUC of 0.924 and an AIC of 186.7. Removing VR from the best model led to a decrease in AUC by 0.073 and an increase in AIC to 205.8, reinforcing its central role in ARDS mortality prediction. To evaluate the clinical utility of the proposed model, we computed detailed performance

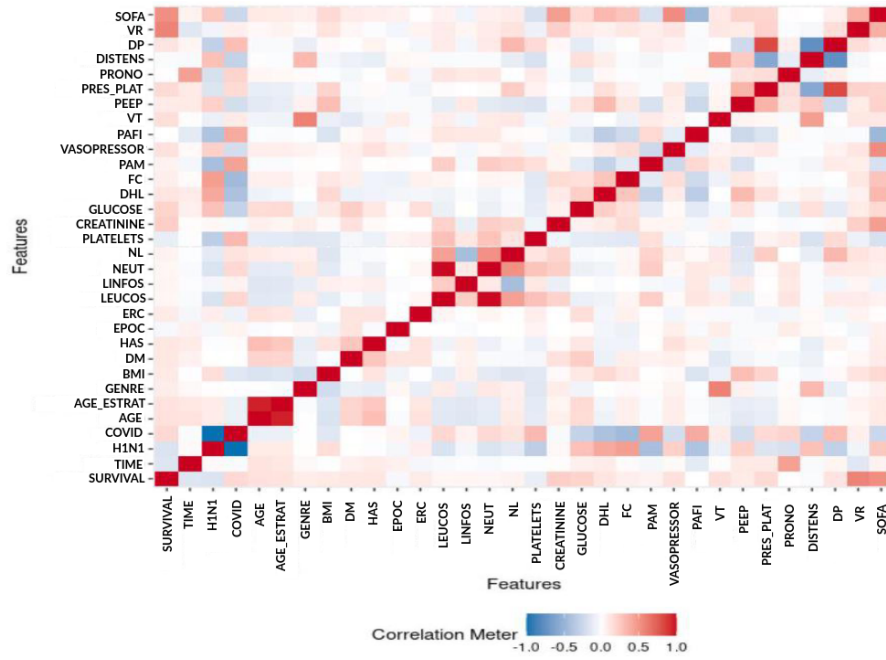


Figure 3. Correlation matrix of dataset

Table 3. Feature importance and its respective coefficient.

Random Forest	Naive Bayes	ANN	MRMR	Chi2	ReliefF	ANOVA	Kruskal-Wallis
VR	100	VR	100	VR	22.8917	VR	VR
CREATININE	19.684	GLUCOSE	0.116527667	DHL	0.0451	CREATININE	GLUCOSE
GLUCOSE	13.731	NL	0.082339546	COVID	0.0417	GLUCOSE	VR
AGE	12.673	DP	0.082074152	DM	0.0292	PRES_PLAT	CREATININE
TIME	12.569	PRES_PLAT	0.081966159	TIME	0.0245	GLUCOSE	PRES_PLAT
PAFI	11.727	HAS	0.064282324	CREATININE	0.019	COVID	TIME
DHL	11.472	CREATININE	0.063112755	GLUCOSE	0.0152	VT	COVID
PRES_PLAT	10.795	DHL	0.044864793	HAS	0.0134	TIME	DHL
DISTENS	9.38	TIME	0.04417583	VT	0.0122	DP	DP
PLATELETS	9.259	PEEP	0.042004424	DHL	0.0119	AGE	HAS
FC	7.505	VR	76.57	PRES_PLAT	0.0106	HAS	VT
NL	6.774	PAM	75.86	GLUCOSE	0.0103	Vasopressor	AGE
VT	6.4	TIME	74.04	BMI	0.0086	DM	PAM
LEUCOS	6.115	Vasopressor	72.29	PRONO	0.0075	PRES_PLAT	Vasopressor
DP	6.084	DM	56.16	GENRE	0.0071	FC	DM
BMI	5.969	AGE	56.01	DP	0.0071	HAS	PAM
PAM	5.357	PRONO	50.08	PAM	0.0071	NL	DHL
PRONO	3.956	COVID	49	PEEP	0.0028	PAFI	PEEP
PEEP	3.256	LEUCOS	48.19	LEUCOS	0.0012	DM	PLATELETS
COVID	3.146	CREATININE	45.23	AGE	-0.0018	GENRE	NL
		FC	38.47	FC	-0.002	PRONO	GENRE
		DISTENS	37.92	PLATELETS	-0.0026	DISTENS	FC
				PAFI	-0.0102	Vasopressor	DISTENS
				DISTENS	-0.01	PAM	LEUCOS
				Vasopressor	-0.0133	VT	PAFI
				NL	0.0711	LEUCOS	PRONO

metrics using 5-fold cross-validation. The model achieved a sensitivity of 87.3%, specificity of 84.2%, positive predictive value (precision) of 85.4%, and negative predictive value of 86.2%. The F1-score was 86.3%, indicating a balanced trade-off between precision and recall. The area under the receiver operating characteristic curve (AUC-ROC) was 0.92, and the area under the precision-recall curve (AUC-PR) was 0.883. These values demonstrate strong discriminative performance. The confusion matrix is presented in Fig. ??, and the precision-recall curves are shown in Fig. ??.

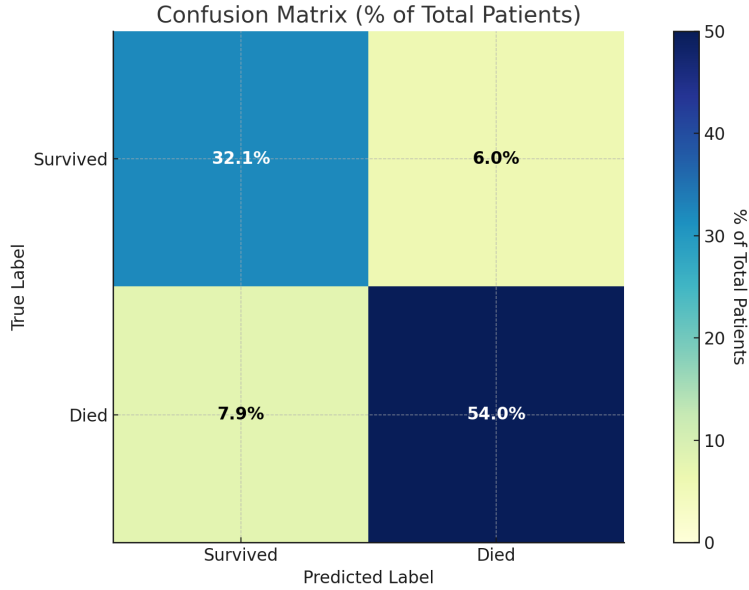


Figure 4. Confusion matrix for the ARDS severity prediction model.

To enhance clinical applicability, the final logistic regression models were stratified by the number of input variables, allowing flexibility depending on available data in real-world ICU settings. Each configuration was calibrated to ensure valid risk probability estimates and supported by internal validation metrics. This scalable modeling approach ensures that clinicians can still obtain reliable mortality predictions even when some variables are missing or delayed. The selection of these logistic regression models, based on varying numbers of variables, highlights the importance of balancing model complexity with clinical applicability. Each model offers unique advantages depending on the clinical scenario and the available data, ensuring that the models can be adapted to different healthcare settings. The use of these models can significantly improve the assessment and management of ARDS patients, providing tailored predictions that align with the specific needs of the clinical environment.

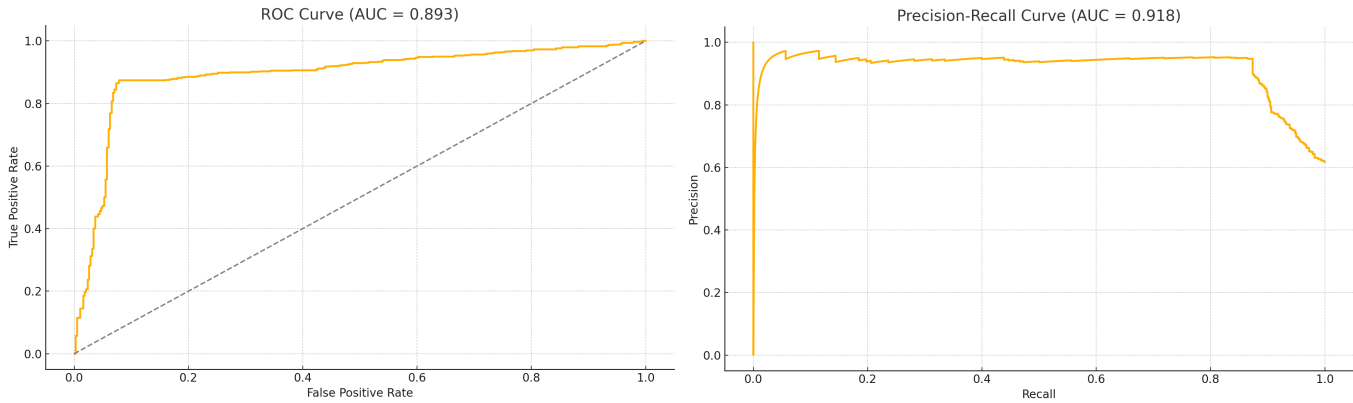


Figure 5. Precision-recall curve with AUC = 0.883, showing the trade-off between sensitivity and precision.

To facilitate the clinical application of our ARDS severity models, we developed a graphical user interface (GUI) using Python’s `tkinter` library. The tool enables clinicians and researchers to interact with the final logistic regression models in

Table 4. Summary of top 5 logistic models with 1 variable

Best 5 Models with 1 variable					
Variables	Estimate	Odds Ratio	McFadden R2	AIC	AUC
VR	-8.562, 3.142	1.9x10 ⁻⁴ , 23.15	0.318	212.016	0.866
Creatinine	-1.291,0.429	0.275,1.536	0.04	296.64	0.635
DHL	-1.193,0.001	0.303,1.001	0.011	305.626	0.608
Time	-0.171,-0.037	0.843,0.964	0.023	301.838	0.609
Pres_plat	-2.581,0.066	0.076,1.069	0.023	302.036	0.606

Table 5. Summary of top 5 logistic models with 3 variables.

Best 5 Models with 3 variables					
Variables	Estimate	Odds Ratio	McFadden R2	AIC	AUC
VR,Covid,Pres_plat	-10.153, 3.113, 0.718, 0.044	3.8x10 ⁻⁵ , 22.492, 2.05, 1.045	0.338	209.723	0.877
VR,Glucose,HAS	-10.412,3.374,0.006, 1.045	3.1x10 ⁻⁵ , 29.2, 1.006, 2.844	0.363	202.183	0.887
VR,Glucose,VT	-12.758, 3.264, 0.007, 0.007	2.8x10 ⁻⁶ , 26.158, 1.007, 1.007	0.36	203.19	0.885
VR,Covid,VT	-11.403, 3.147,0.737, 0.006	1.1x10 ⁻⁵ , 23.258, 2.089, 1.006	0.344	207.875	0.874
VR,Glucose,Time	-9.321, 3.181, 0.007, -0.033	8.9x10 ⁻⁵ , 24.063, 1.007, 0.968	0.356	204.494	0.884

Table 6. Summary of top 5 logistic models with 4 variables.

Best 5 Models with 4 variables					
Variables	Estimate	Odds Ratio	McFadden R2	AIC	AUC
VR,Covid,Glucose, DHL	-12.012, 3.169, 1.742, 0.009, 0.001	6x10 ⁻⁶ , 23.77, 5.709, 1.009,1.001	0.397	193.733	0.905
VR,Covid,HAS, Glucose	-11.8, 3.376, 1.374, 1.075, 0.01	7.5x10 ⁻⁶ , 29.254,3.95, 2.93, 1.01	0.4	193.091	0.906
VR,Covid,Glucose, Age	-12.76, 3.237, 1.494, 0.009, 0.033	2.8x10 ⁻⁶ , 25.462, 4.456, 1.009, 1.003	0.394	194.733	0.902
VR,Covid,Glucose, Vasopressor	-11.426, 3.129,0.01, 1.534, 0.726	1.09x10 ⁻⁶ , 22.858,1.01, 4.635, 2.067	0.393	195.062	0.905
VR,Covid,VT, Glucose	-13.652,3.236, 1.295,0.006, 0.01	1.1x10 ⁻⁶ , 25.437, 3.651, 1.006, 1.01	0.393	195.173	0.905

Table 7. Summary of top 5 logistic models with 5 variables.

Best 5 Models with 5 variables					
Variables	Estimate	Odds Ratio	McFadden R2	AIC	AUC
VR,DHL,Covid, Glucose, Age	-14.72, 3.195, 0.002, 2.118, 0.008, 0.047	4x10 ⁻⁷ , 24.408, 1.002, 8.315, 1.008, 1.048	0.42	188.859	0.91
VR,Covid,HAS, DHL, Glucose	-12.822, 3.33, 1.821,1.18, 0.002,0.009	2.7x10 ⁻⁶ , 27.932,6.181,3.253,1.002,1.009	0.419	189.267	0.911
VR,DHL,Covid Glucose, Vasopressor	-12.487, 3.091, 0.002, 1.992, 0.009, 0.806	3.7x10 ⁻⁶ , 22, 1.002, 7.329, 1.009, 2.239	0.412	191.44	0.911
VR,Covid,HAS, VT, Glucose	-14.251,3.417,1.344,1.153,0.007,0.01	4.3x10 ⁻⁷ , 30.474,3.834,3.169,1.007,1.01	0.413	190.922	0.91
VR,Covid,Time, HAS, Glucose	-11.211,3.324,1.353,-0.039,1.193,0.01	1.3x10 ⁻⁵ , 27.77, 3.867, 0.962, 3.298, 1.01	0.413	191.021	0.906

Table 8. Summary of top 5 logistic models with 6 variables.

Best 5 Models with 6 variables					
Variables	Estimate	Odds Ratio	McFadden R2	AIC	AUC
VR,Glucose,Time, DHL,Age,Covid	-14.403, 3.11, 0.009, -0.044, 0.002,0.053, 2.196	5.5x10-7, 22.412, 1.009, 0.957, 1.002, 1.054, 8.985	0.436	185.861	0.914
VR,DHL,Covid, Glucose,Vasopressor, Age	-15.248, 3.107, 0.002, 2.39, 0.009, 0.815, 0.047	2.3x10-7, 22.353, 1.002, 10.916, 1.009, 2.259, 1.048	0.434	186.677	0.916
VR,Covid,Time, HAS,DHL,Glucose	-12.333,3.291,1.853,-0.045,1.341,0.002,0.009	4.4x10-6,26.867,6.38,0.956,3.822,1.002,1.009	0.436	186.102	0.914
VR,Covid,HAS, VT,DHL,Glucose	-16.085,3.405,1.806,1.266,0.007,0.002,0.009	1.1x10-7, 30.121,6.085,3.546,1.007,1.002,1.009	0.434	186.576	0.916
VR,DHL,Covid, Glucose,HAS,Age	-15.018,3.339,0.002,2.138,0.009,0.968,0.039	3x10-7,28.184,1.002,8.481,1.009,2.633,1.04	0.433	186.865	0.914

Table 9. Summary of top 5 logistic models with 7 variables.

Best 5 Models with 7 variables					
Variables	Estimate	Odds Ratio	McFadden R2	AIC	AUC
VR,Covid,Time, HAS,VT,DHL, Glucose	-15.749,3.377,1.83,-0.046,1.433,0.008,0.002,0.009	1.4x10-7,29.281,6.232,0.955,4.192,1.008,1.002,1.009	0.452	183.093	0.919
VR,Glucose,Time, DHL,Age,Platelets, Covid	-13.79, 3.075,0.009, -0.044, 0.002, 0.052, -0.004,2.556	1x10-6, 21.64, 1.009, 0.957, 1.002, 1.053, 0.996, 12.889	0.447	184.594	0.916
VR,Glucose,Time, DHL,Age,PEEP, Covid	-15.578, 3.093, 0.009, -0.048, 0.002, 0.058, 0.088, 2.33	1.7x10-7, 22.05, 1.009, 0.953, 1.002, 1.06, 1.092, 10.279	0.443	185.87	0.915
VR,DHL,Covid, Glucose,HAS, Vasopressor,Age	-15.528,3.252,0.002,2.403,0.009,0.888,0.748,0.04	1.8x10-7,25.842,1.002,11.054,1.009,2.43,2.113,1.041	0.444	185.451	0.918
Creatinine,VR,Glucose, Time,DHL,Age, Covid	-14.343, 0.21, 3.023, 0.009, -0.043, 0.002, 0.05, 2.214	5.9x10-7, 1.233, 20.557, 1.009, 0.958, 1.002, 1.052, 9.154	0.442	186.964	0.916

Table 10. Summary of top 5 logistic models with 8 variables.

Best 5 Models with 8 variables					
Variables	Estimate	Odds Ratio	McFadden R2	AIC	AUC
VR,Covid, Time,Creatinine,HAS, VT,DHL,Glucose	-15.984,3.303,1.902,-0.045,0.233,1.424,0.008,0.002,0.009	1.1x10-7,27.198,6.696,0.956,1.262,4.152,1.008,1.002,1.009	0.46	182.764	0.923
VR,Covid,Time, Creatinine,HAS,Pres_plat, DHL,Glucose	-13.089,3.198,1.852,-0.046,0.233,1.274,0.027,0.002,0.009	2x10-6,24.476,6.375,0.955,1.262,3.575,1.028,1.002,1.01	0.445	183.31	0.919
Creatinine,VR,Glucose, Time,DHL,Age, Platelets,Covid	-13.745,0.21,2.987,0.009,-0.044,0.002,0.05,-0.004,2.593	1x10-6,1.234,19.816,1.009,0.957,1.002,1.051,0.996,13.367	0.453	184.762	0.92
VR,Glucose,Creatinine, Time,Covid,DHL, HAS,DP	,-12.487,3.216,0.009,0.233,-0.044,1.905,0.002,1.309,0.001	3.7x10-6,24.934,1.009,1.262,0.957,6.719,1.002,3.702,1.001	0.443	185.831	0.919
VR,Covid, Time,HAS,DM, DHL,Pres_plat,Glucose	-12.991,3.277,1.807,-0.047,1.202,0.453,0.002,0.026,0.009	2.2x10-6,26.508,6.093,0.954,3.326,1.574,1.002,1.027,1.009	0.44	186.752	0.917

Table 11. Summary of top 5 logistic models with 9 variables.

Best 5 Models with 9 variables					
Variables	Estimate	Odds Ratio	McFadden R2	AIC	AUC
VR,Covid,DM, Time,Creatinine,HAS, VT,DHL,Glucose	-16.224, 3.311,1.914,0.505,-0.047,0.196,1.314,0.008,0.002,0.009	9x10-8, 27.408,6.78,1.657,0.954,1.216,3.722,1.008,1.002,1.009	0.463	183.772	0.923
VR,Covid,Time, Creatinine,HAS,VT, DHL,Pres_plat,Glucose	-16.459,3.289,1.85,-0.047,0.232,1.386,0.008,0.002,0.023,0.009	7.1x10-8, 26.816,6.358,0.954,1.261,3.998,1.008,1.002,1.023,1.009	0.461	184.412	0.924
Creatinine,VR,Glucose, Time,DHL,Age, Platelets,PEEP,Covid	-14.981, 0.233, 2.96, 0.009, -0.048, 0.002, 0.055, -0.003, 0.086, 2.701	3.1x10-7, 1.249, 19.296, 1.009, 0.953, 1.002, 1.057, 0.997, 1.09, 14.895	0.459	184.974	0.922
VR,Glucose,Creatinine, Time,Covid,DHL, DP,HAS,VT	-15.969,3.303,0.009,0.233,-0.045,1.906,0.002,-0.001,1.426,0.008	1.2x10-7, 27.202,1.009,1.262,0.956,6.726,1.002,0.999,4.162,1.008	0.46	184.763	0.924
VR,Covid,DM, Time,HAS,VT, DHL,Pres_plat,Glucose	-16.514,3.37,1.794,0.621,-0.048,1.262,0.008,0.002,0.021,0.009	6.7x10-8, 29.082,6.013,1.86,0.953,3.532,1.008,1.002,1.002,1.009	0.458	185.218	0.921

Table 12. Summary of top 5 logistic models with 10 variables.

Best 5 Models with 10 variables					
Variables	Estimate	Odds Ratio	McFadden R ²	AIC	AUC
VR,Covid,DM,Time,Creatinine,HAS,VT,DHL,Pres_plat,Glucose	-16.681,3.295,1.863,0.502,-0.048,0.196,1.276,0.008,0.002,0.023,0.009	5.6x10 ⁻⁸ , 26.986,6.441,1.652,0.953,1.216,3.581,1.008,1.002,1.023,1.009	0.464	185.437	0.924
VR,Glucose,Creatinine,Pres_plat,Time,Covid,DHL,DP,HAS,VT	-16.692,3.271,0.009,0.243,0.068,-0.048,1.966,0.002,-0.061,1.424,0.008	5.6x10 ⁻⁸ ,26.35,1.009,1.275,1.071,0.953,7.143,1.002,0.941,4.154,1.008	0.463	185.683	0.925
Creatinine,VR,Glucose,Time,DHL,Age,Platelets,PEEP,Covid,BMI	-15.784, 0.212, 2.952, 0.009, -0.046, 0.002, 0.058, -0.003, 0.08, 2.729, 0.021	1.3x10 ⁻⁷ , 1.236, 19.141, 1.009, 0.955, 1.002, 1.06, 0.997, 1.083, 15.316, 1.021	0.46	186.505	0.923
VR,Creatinine,Pres_plat,Glucose,Covid,VT,Time,DP,Age,HAS	-18.107,3.245,0.266,0.128,0.01,1.669,0.008,-0.049,-0.105,0.038,1.051	1.3x10 ⁻⁸ , 25.673,1.305,1.136,1.01,5.305,1.008,0.952,0.9,1.039,2.86	0.458	187.273	0.921
VR,DHL,Covid,Pres_plat,Creatinine,Glucose,HAS,Vasopressor,FC,Age	-16.783,3.12,0.002,2.538,0.017,0.195,0.009,0.853,0.731,0.01,0.039	5.1x10 ⁻⁸ ,22.64,1.002,12.656,1.017,1.215,1.009,2.346,2.077,1.01,1.04	0.453	188.702	0.922

an intuitive and efficient manner. The interface was designed to support early risk stratification in ICU settings by allowing users to input patient-specific variables and receive real-time mortality probability estimates based on the selected model as shown in Fig. ???. The GUI offers ten logistic regression models, each derived from combinations of the top-ranked variables identified during the feature selection phase. These models vary in complexity, ranging from a single-variable model based solely on ventilatory ratio (VR) to a ten-variable model that incorporates VR, glucose, age, time since symptom onset, DHL, COVID-19 status, platelet count, creatinine, PEEP, and BMI. This design allows the user to select a model best suited to the available clinical data at the time of admission, making the tool robust to partial data scenarios commonly encountered in the ICU. Upon selecting a model from a drop-down menu, the interface dynamically generates the corresponding variable input fields. After entering the values, the user can compute the mortality probability with a single click. Internally, the tool applies the corresponding logistic regression equation using the model coefficients established during training and validated in our analysis (coefficients referenced from Table 4-12). The probability is then displayed in a popup message, providing immediate and interpretable feedback.

This GUI supports decision-making at the bedside by offering an explainable, flexible, and validated interface for early mortality prediction. Its modular design enables seamless updates or extensions with additional models, datasets, or predictors as needed. The source code and installation instructions are provided in the accompanying supplementary repository to facilitate reproducibility and institutional deployment.

Conclusion

This supplementary material reinforces the methodological rigor and clinical relevance of the proposed ARDS severity prediction framework. Through a combination of statistical, information-theoretic, and machine learning-based feature selection techniques, we identified a consistent subset of physiologically interpretable predictors—most notably the ventilatory ratio (VR), glucose, creatinine, and COVID-19 status—that demonstrated strong and reproducible associations with mortality outcomes. The logistic regression models developed using these selected features showed excellent predictive performance, with the top models achieving an AUC-ROC of 0.924 and outperforming the traditional SOFA score. Feature selection was performed within a nested cross-validation scheme to minimize bias and ensure generalizability. Additionally, the exhaustive model evaluation based on AIC permitted the selection of models that balance accuracy, parsimony, and clinical interpretability. Importantly, our results demonstrate that VR is the most robust and discriminative predictor across all methods. Its inclusion significantly improved model fit and discrimination. The modular structure of the final models, allowing use of 1 to 10 variables, supports flexible deployment in real-world ICU settings, where data availability may vary across patients and time. Together, these findings support the implementation of a transparent, scalable, and interpretable scoring tool for early mortality risk stratification in ARDS patients, offering a valuable complement to existing severity scores and aiding evidence-based clinical decision-making.

Author contributions statement

CHC, GLG, JCA were involved in the study design, acquisition of data, manuscript preparation and final approval of the work. GLT, MARA, IMB were involved in the analysis and interpretation of data and manuscript preparation and final approval of the work.

Data availability statement

All patient data were anonymized and handled in accordance with ethical guidelines for clinical research. The study was conducted with approval from the institutional review board with protocol code number C26-24, ensuring compliance with ethical standards in data handling and analysis. The dataset used in this study contains sensitive patient data and is not publicly available. Researchers interested in accessing the dataset may contact the corresponding author upon reasonable request and subject to ethical approval.

Additional information

The most relevant results for data processing, alternative propose models and mortality ARDS calculator code can be found at: <https://github.com/Lugo1025/ARDS>.

Competing interests

The author(s) declare no competing interests.

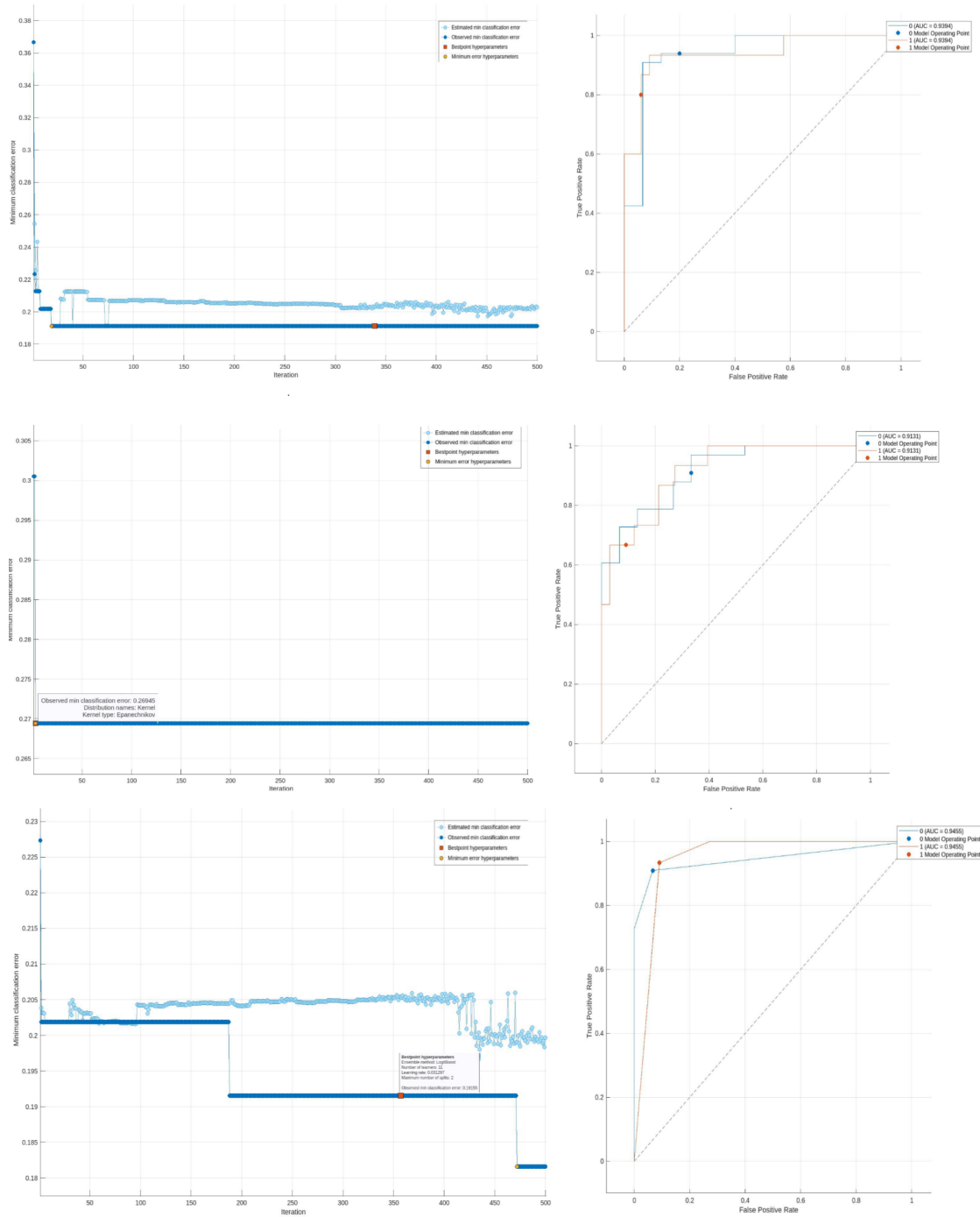


Figure 6. Training and performance evaluation of the three wrapper-based classifiers used for feature selection. From top to bottom: Random Forest, Naive Bayes, and Artificial Neural Network (ANN). For each model, hyperparameter optimization was performed using a Bayesian strategy with 5-fold cross-validation and up to 500 iterations. Performance is reported in terms of minimum classification error and AUC-ROC, supporting the robustness of feature importance extraction.

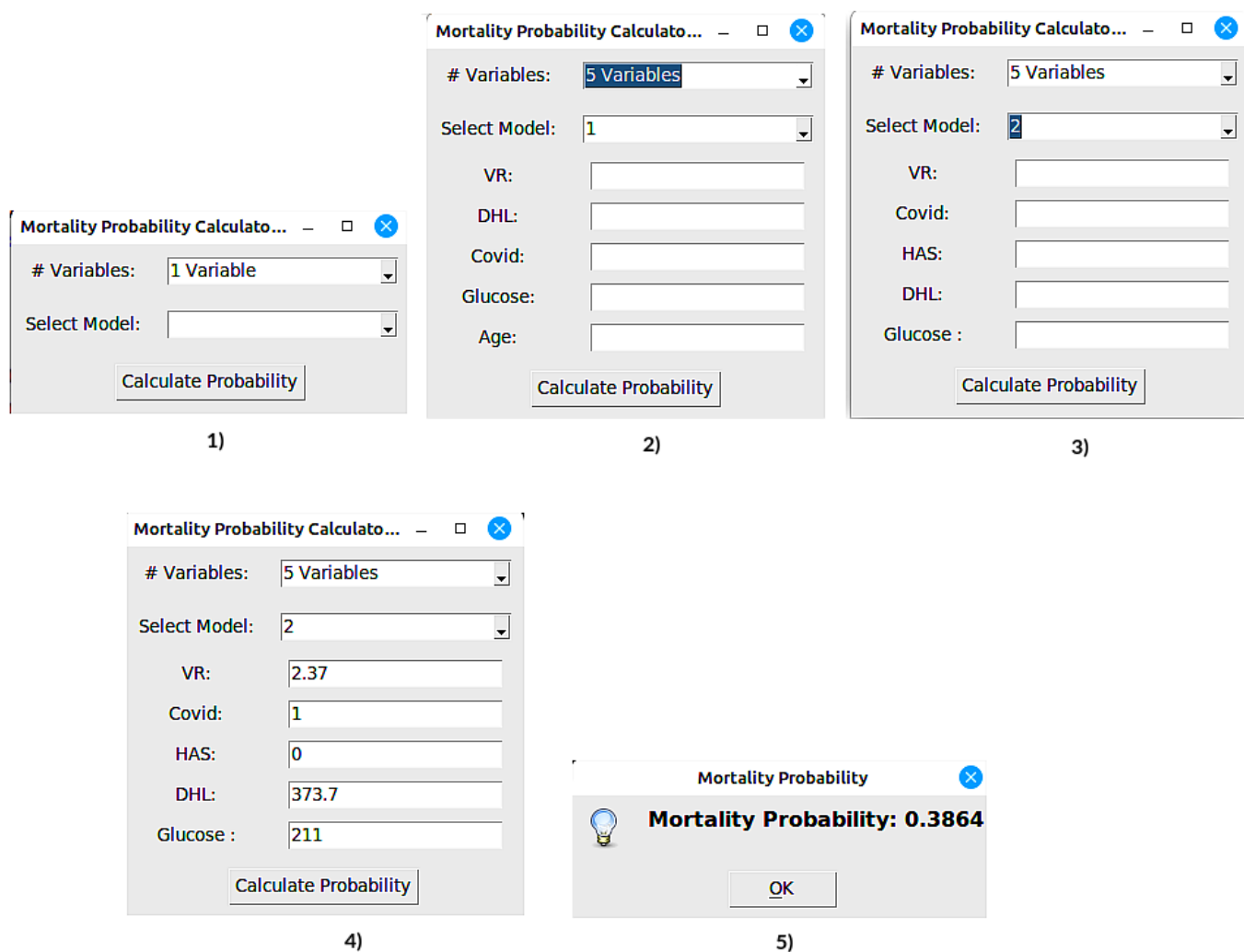


Figure 7. Graphical user interface (GUI) for ARDS mortality prediction. From right to left: (1) Initial window; (2) Selection of number of variables (1–10); (3) Selection of model index (1–5); (4) Input of patient variables; (5) Display of predicted mortality. Example shown: a patient with VR = 2.37, COVID-19 positive, no HAS, DHL = 373.7 U/dL, glucose = 211 mg/dL yields a 38.64% mortality risk, consistent with survival. The tool supports rapid bedside use and is available in the repository listed under Additional Information.